

Brain Tumor MRI Classification

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Abstract—This report establishes the methodological foundation for a patient-independent, explainable, and computationally efficient multiclass brain tumor MRI classification system. The planned task is to classify anonymised MRI slices into four categories glioma, meningioma, pituitary tumor, and no tumor using publicly available datasets and models that are practical to train and deploy under limited computational resources. We critically review ten core studies and one methodological paper across three themes: transfer-learning performance, region-aware analysis and explainability, and evaluation reliability. The review indicates that reported performance is strongly influenced by data-splitting procedures, class imbalance, and the limited validation of model explanations. In response, we define a leakage-resistant evaluation protocol, class-sensitive metrics, lightweight transfer-learning baselines, and a controlled comparison of image preprocessing and feature representations. Grad-CAM will be used as a diagnostic explanation method and evaluated quantitatively when suitable tumor-region annotations are available. The resulting framework prioritises a careful balance among predictive performance, computational cost, generalisation to unseen patients, and interpretability. The study is designed to provide a reproducible and resource-conscious evaluation rather than to maximise complexity or benchmark accuracy alone.

Index Terms—Brain Tumor Classification, Magnetic Resonance Imaging, Convolutional Neural Networks, Transfer Learning, Explainable AI, Data Leakage, Patient-Independent Validation, Grad-CAM.

I. INTRODUCTION

Automated MRI-based diagnosis is one of the most heavily benchmarked problems in medical imaging, yet reported accuracies routinely exceed 98% while clinical adoption remains limited.

Three structural challenges distinguish this problem from a generic image-classification benchmark. First, public brain MRI datasets are small relative to natural-image corpora, and multiple slices are frequently drawn from the same patient a correlation structure that standard random image-level splitting ignores, precisely the failure mode formalised by Yagis et al. [11]. Second, four tumor classes (glioma, meningioma, pituitary, no-tumor) are rarely balanced, so accuracy alone the metric most commonly reported across the reviewed literature [3, 10] can mask poor minority-class recall. Third, the field’s dominant interpretability tool, Grad-CAM, is almost universally displayed but rarely validated quantitatively against ground-truth tumor regions [6, 7], leaving open whether high-accuracy models are actually attending to clinically meaningful anatomy.

II. LITERATURE REVIEW

Ten core papers and one essential methodological paper are reviewed and organised into three thematic clusters. Rather

than cataloguing architectures, this review constructs a critical argument: the field’s most consistent unaddressed flaw is the conflation of in-dataset accuracy with genuine patient-level generalisation. Every modelling decision in Sections V–VI is derived from this body of work.

A. Cluster I: Transfer-Learning Benchmarks and the Accuracy Ceiling

Deepak and Ameer (2019) [9] were among the first to apply pretrained GoogLeNet features to three-class tumor classification, explicitly using patient-level five-fold cross-validation a methodological choice that remains uncommon in later work. Disci et al. (2025) [3] systematically compare six pretrained CNNs (VGG, ResNet, Xception, EfficientNet variants) on a 7,023-image four-class dataset, reporting that Xception achieves the strongest weighted accuracy and weighted F1-score. Vimala et al. (2023) [7] compare EfficientNetB0–B7 on three-class contrast-enhanced MRI with augmentation and Grad-CAM, identifying EfficientNetB0 as the strongest lightweight candidate. Ilani, Shi, and Banad (2025) [2] extend this cluster with explicit external cross-dataset validation using U-Net, CNN, InceptionV3, EfficientNetB0, and VGG16.

Our synthesis: transfer learning reliably compresses training cost while achieving benchmark accuracies above 98%, but architecture choice alone is not a meaningful research contribution these four studies differ mainly in preprocessing, class balance, and splitting protocol, not in genuine methodological novelty. This is the primary justification for treating MobileNetV2/EfficientNetB0 as fixed lightweight baselines rather than as the object of study.

B. Cluster II: Localisation, Feature Fusion, and Interpretability

Aamir et al. (2025) [1] combine enhancement, ROI extraction, attention, and ensemble classification on BraTS2020 and Figshare, reporting very high accuracy but at substantial architectural and computational cost. Kathuria, Gupta, and Uppal (2025) [4] propose a unified CNN framework for joint classification and segmentation. Iqbal et al. (2024) [14] fuse deep spatial features with handcrafted radiomic descriptors (“FusionNet”), reporting an F1-score of 96.12 ± 0.41 on a binary task across BraTS, TCIA, and Figshare. Jafari et al. (2025) [15] show that GLCM/Curvelet radiomics with classical classifiers (Random Forest, CatBoost) reach 95.2% accuracy without any deep feature learning. Musthafa et al. (2024) [6] and Iftikhar et al. (2025) [13] pair ResNet50 and a compact custom CNN, respectively, with Grad-CAM; Iftikhar

et al. additionally validate with SHAP and LIME and argue that reducing network depth improves practical deployability.

A tension is worth making explicit: segmentation- and fusion-based methods (Aamir, Kathuria, Iqbal) consistently report accuracy gains attributed to explicit tumor-region information, while the radiomics-only study (Jafari) achieves comparable accuracy with far lower computational cost, and the explainability-focused study (Iftikhar) argues smaller models generalise better in practice. Our synthesis: the incremental value of full segmentation over a simple tumor-centred crop is rarely isolated experimentally in this cluster, and a controlled ablation raw image vs. crop vs. radiomic features is a more resource-efficient way to test the underlying hypothesis than building a full segmentation-and-ensemble pipeline.

C. Cluster III: Evaluation Rigour and Generalisation

Yagis et al. (2021) [11] not a brain tumor study, but the single most load-bearing methodological reference in this review quantitatively demonstrate that slice-level (rather than subject-level) splitting of 2D MRI data can severely inflate reported classification performance, and that augmenting before the train/test split places near-duplicate images on both sides of the split. Ilani et al. [2] independently reinforce this cluster through explicit cross-dataset testing, a stronger generalisation test than the single-dataset protocol used by most of the remaining reviewed studies.

This cluster directly exposes the central weakness running through Clusters I and II: high in-dataset accuracy figures reported by Aamir [1], Disci [3], and others are not necessarily comparable, because it is frequently unclear whether patient identifiers were used for splitting at all. Where identifiers are unavailable, Yagis et al. [11] motivate treating image-level evaluation as an explicit, disclosed limitation rather than presenting it as patient-level validation.

D. Literature-to-Project Design Decision Derivation

Table I makes the literature-to-implementation chain explicit.

III. PROBLEM FORMULATION

The task is patient-independent multiclass image classification. Given an MRI slice $x_i \in \mathbb{R}^{H \times W \times C}$ drawn from patient p_i , the goal is to predict a class label $y_i \in \{\text{glioma, meningioma, pituitary, no-tumor}\}$. Unlike a generic image-classification benchmark, samples are not exchangeable: multiple x_i share the same p_i , so any evaluation split $\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{test}}$ must enforce

$$\{p_i : x_i \in \mathcal{D}_{\text{train}}\} \cap \{p_i : x_i \in \mathcal{D}_{\text{test}}\} = \emptyset, \quad (1)$$

following directly from the leakage mechanism documented by Yagis et al. [11]. Because class frequencies are imbalanced, plain accuracy is an insufficient primary metric; the project instead reports macro-F1,

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{2 \text{Prec}_k \cdot \text{Rec}_k}{\text{Prec}_k + \text{Rec}_k}, \quad (2)$$

TABLE I
LITERATURE-TO-PROJECT DESIGN DECISION DERIVATION

Paper	Key Finding	Phase 1 Decision	Sec.
Yagis [11]	Slice-level leakage inflates accuracy	Patient-level grouped split	VI
Deepak [9]	Patient-level CV benchmark	Baseline patient-level 5-fold CV	VI
Disci [3]	Transfer-learning ceiling; Xception best	MobileNetV2/EfficientNetB0 baseline	VI
Ilani [2]	External validation improves robustness	Held-out external test set	VI
Aamir [1]	ROI + attention boosts accuracy	Tumor-crop ablation arm	VI
Iqbal [14]	Feature fusion competitive with deep nets	Hybrid radiomics-CNN ablation	VI
Jafari [15]	Radiomics + classical ML competitive	Radiomics baseline arm	VI
Musthafa [6]	Grad-CAM shows attention regions	Heatmap generation protocol	VI
Iftikhar [13]	Smaller models deploy better; multi-XAI	Lightweight backbone; SHAP audit	VI
Vimala [7]	EfficientNetB0 strongest lightweight net	Primary deployable model	VI

alongside balanced accuracy, per-class recall, specificity, and one-vs-rest ROC-AUC, with $K = 4$ classes. Grad-CAM outputs are treated as a diagnostic layer rather than as part of the primary metric, in line with the interpretability-validity concerns raised in Cluster II.

IV. PRELIMINARY DATASET CHARACTERISATION

The primary dataset under consideration is the 7,023-image four-class Brain Tumor MRI dataset used by Disci et al. [3], spanning glioma, meningioma, pituitary tumor, and no-tumor classes. Two structural properties inherited from the literature directly shape the pipeline design:

A. Absence of Reliable Patient Identifiers

Several public brain-tumor collections, including the dataset above, do not consistently expose patient-level identifiers. This is the same limitation Yagis et al. [11] warn against: without subject IDs, image-level evaluation risks silently reporting inflated, leakage-contaminated accuracy. The project will explicitly document this limitation and, where identifiers are unavailable, will split *before* augmentation and remove duplicate or near-duplicate images to reduce (though not eliminate) leakage risk.

B. Class Imbalance Across the Four-Class Taxonomy

Consistent with Disci et al.'s [3] and Rasa et al.'s [10] reporting of inconsistent per-class recall despite high overall accuracy, the four tumor classes are not expected to be evenly represented. This motivates the class-weighted loss and macro-averaged metrics defined in Eq. 2, rather than raw accuracy, as the project's primary reported outcome.

V. THEORETICAL RIGOUR AND METHODOLOGICAL REQUIREMENTS

Applying standard image-classification practice to this task blindly violates the patient-independence assumption implicit in most benchmark accuracy figures. This section links the cluster-level findings above to concrete methodological constraints.

A. Patient-Level Independence Violations

Random image-level k -fold splitting, still common in the reviewed literature [1, 3], permits correlated slices from the same patient to appear on both sides of a split, directly contradicting Eq. 1 and inflating apparent generalisation, as formalised by Yagis et al. [11]. The project mandates grouped, patient-level (or, where unavailable, pre-augmentation image-level) splitting for every reported result.

B. Metric Sensitivity Under Class Imbalance

Because overall accuracy is insensitive to per-class failure, and the reviewed literature [3, 10] shows this can mask weak minority-class recall, macro-F1 and balanced accuracy (Eq. 2) are treated as the primary reported metrics, with accuracy retained only as a secondary, comparison-friendly figure.

C. Explainability Validity

Grad-CAM visualisations are widely reported [6, 7] but rarely validated quantitatively. Following this gap, the project will, wherever segmentation masks or expert region annotations are available, compute heatmap-mask overlap rather than relying on visual inspection alone, treating Grad-CAM as a diagnostic tool rather than proof of clinical validity.

VI. PROPOSED ARCHITECTURE AND PREPROCESSING PIPELINE

To respect the theoretical constraints above and the literature-grounded decisions in Table I, the Phase 1 design specifies the following:

- 1) **Patient-Level (or Leakage-Resistant) Splitting.** All data is partitioned according to Eq. 1 where subject identifiers exist; where they do not, splitting precedes augmentation and near-duplicates are removed. This implements the central corrective motivated by Yagis et al. [11].
- 2) **Class-Balanced Lightweight Transfer Learning.** A simple two-to-three-block CNN serves as the from-scratch baseline; MobileNetV2 and EfficientNetB0 identified as the strongest lightweight transfer-learning candidates in Cluster I [3, 7] serve as the primary deployable models, trained with class-weighted cross-entropy to address the imbalance documented in Section IV-B.
- 3) **Tumor-Region Ablation.** Following the tension identified in Cluster II, the pipeline compares the raw image, a tumor-centred crop, and a radiomics-plus-classical-ML arm [14, 15] under an identical evaluation protocol, isolating the marginal value of explicit localisation without committing to a full segmentation network.

- 4) **Quantitatively Audited Explainability.** Grad-CAM is generated for every model per Cluster II [6, 13], with heatmap-mask overlap reported wherever ground-truth regions are available, directly addressing the interpretability-validity gap in Section V-C.
- 5) **Robustness and Reproducibility.** Following the field-wide absence of controlled robustness testing, held-out test images are additionally evaluated under mild Gaussian noise, contrast shift, and small rotations without retraining; all splits, seeds, and preprocessing parameters are published for reproducibility.

VII. CONCLUSION

Phase 1 successfully scopes the methodological hostility underlying an apparently “solved” benchmark problem. We have shown that headline accuracies above 98% across the reviewed literature are frequently reported without patient-level splitting [11], without external cross-dataset validation [2], without class-sensitive metrics [3, 10], and without quantitatively validated explainability [6, 7]. Each Phase 1 design decision is derived from a specific finding in the reviewed literature rather than adopted as a default heuristic.

The central argument, grounded in the literature review, is that this benchmark problem’s greatest remaining challenge is not architectural but evaluative: genuine patient-level generalisation, honestly measured, rather than marginal accuracy gains from deeper or more complex networks.

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